

Attention

- Selecting things to concentrate on at a point in time from the mass of stimuli around us
- Allows us to focus on information that is relevant to what we are doing
- Involves audio and/or visual senses
- Focussed and divided attention enables us to be selective in terms of the mass of competing stimuli but limits our ability to keep track of all events
- Information at the interface should be structured to capture users' attention, e.g. use perceptual boundaries (windows), colour, reverse video, sound and flashing lights

Activity: Find the price of a double room at the Holiday Inn in Columbia

South Carolina					
City	Motel/Hotel	Area code	Phone	Rates	
				Single	Double
Charleston	Best Western	803	747-0961	\$126	\$130
Charleston	Days Inn	803	881-1000	\$118	\$124
Charleston	Holiday Inn N	803	744-1621	\$136	\$146
Charleston	Holiday Inn SW	803	556-7100	\$133	\$147
Charleston	Howard Johnsons	803	524-4148	\$131	\$136
Charleston	Ramada Inn	803	774-8281	\$133	\$140
Charleston	Sheraton Inn	803	744-2401	\$134	\$142
Columbia	Best Western	803	796-9400	\$129	\$134
Columbia	Carolina Inn	803	799-8200	\$142	\$148
Columbia	Days Inn	803	736-0000	\$123	\$127
Columbia	Holiday Inn NW	803	794-9440	\$132	\$139
Columbia	Howard Johnsons	803	772-7200	\$125	\$127
Columbia	Quality Inn	803	772-0270	\$134	\$141
Columbia	Ramada Inn	803	796-2700	\$136	\$144
Columbia	Vagabond Inn	803	796-6240	\$127	\$130

Activity: Find the price for a double room at the Quality Inn in Pennsylvania

Pennsylvania
Bedford Motel/Hotel: Crinaline Courts
(814) 623-9511 S: \$118 D: \$120
Bedford Motel/Hotel: Holiday Inn
(814) 623-9006 S: \$129 D: \$136
Bedford Motel/Hotel: Midway
(814) 623-8107 S: \$121 D: \$126
Bedford Motel/Hotel: Penn Manor
(814) 623-8177 S: \$119 D: \$125
Bedford Motel/Hotel: Quality Inn
(814) 623-5189 S: \$123 D: \$128
Bedford Motel/Hotel: Terrace
(814) 623-5111 S: \$122 D: \$124
Bradley Motel/Hotel: De Soto
(814) 362-3567 S: \$120 D: \$124
Bradley Motel/Hotel: Holiday House
(814) 362-4511 S: \$122 D: \$125
Bradley Motel/Hotel: Holiday Inn
(814) 362-4501 S: \$132 D: \$140
Breezewood Motel/Hotel: Best Western Plaza
(814) 735-4352 S: \$120 D: \$127
Breezewood Motel/Hotel: Motel 70
(814) 735-4385 S: \$116 D: \$118

Activity

- Tullis (1987) found that the two screens produced quite different results
 - 2nd screen - took an average of 5.5 seconds to search
 - 1st screen - took 3.2 seconds to search
- Why, since both displays have the same density of information (31%)?
- Spacing
 - In the 2nd screen the information is bunched up together, making it hard to search
 - In the 1st screen the characters are grouped into vertical categories of information making it easier

Multitasking and attention

- Is it possible to perform multiple tasks without one or more of them being detrimentally affected?
- Ophir et al (2009) compared heavy vs light multi-taskers
 - heavy were more prone to being distracted than those who infrequently multitask
 - heavy multi-taskers are easily distracted and find it difficult to filter irrelevant information

Design implications for attention

- Make information salient when it needs attending to
- Use techniques that make things stand out like color, ordering, spacing, underlining, sequencing and animation
- Avoid cluttering the interface with too much information
- Search engines and form fill-ins that have simple and clean interfaces are easier to use